AI Layoff Risk Predictor

AI101 Mini-Project | Team 28

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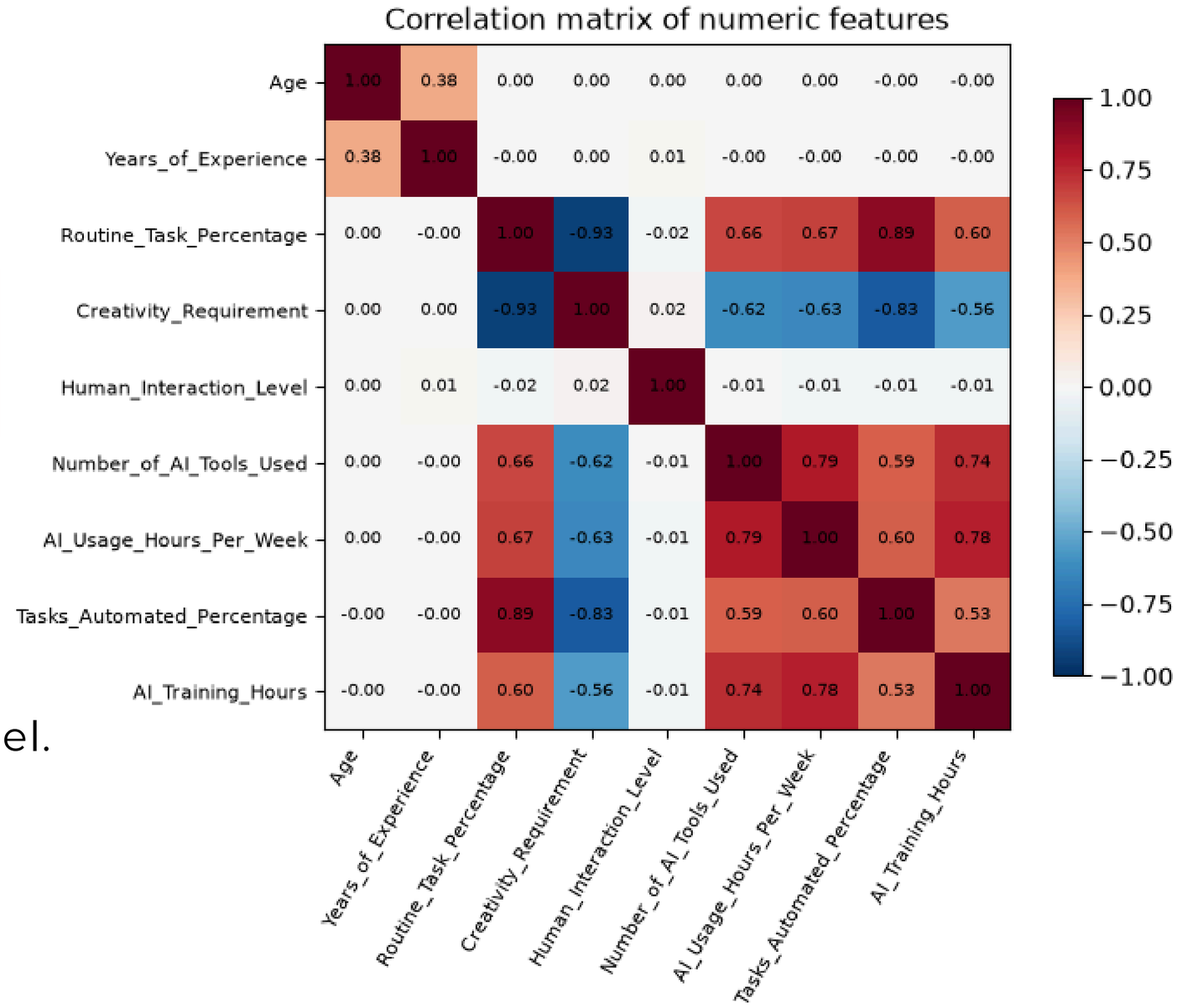
Repo Link: https://github.com/Shreyas-IITRPR/AI-Layoff-Risk-Predictor

Problem and data:

20,000 employee records

20,000	16	3
Rows	columns	risk classes

What and Why: Predict layoff risk as AI adoption grows.
Source: Kaggle, AI Impact on Jobs and Layoff Risk (synthetic).
Columns: 9 numeric, 6 categorical, 1 label.
Statistics: Risk rises with routine tasks (r = 0.78), falls with creativity (r = -0.75).



Dataset Link: https://www.kaggle.com/datasets/shivasingh4945/ai-impact-on-jobs-and-layoff-risk-dataset?hl=en-US

Approach: One Logistic Regression Pipeline

1
DATA
20,000-row CSV in the repo

2
PREPARE
Check missing and duplicates, 80/20 stratified split, scale + one-hot (54 features)

3
TRAIN
Logistic Regression ; SGD loss curve over 100 epochs

4
EVALUATE
Train and test: confusion matrix, ROC, accuracy, precision, recall, F1

5
OUTPUT
Notebook, plots, metrics.json

Libraries & versions

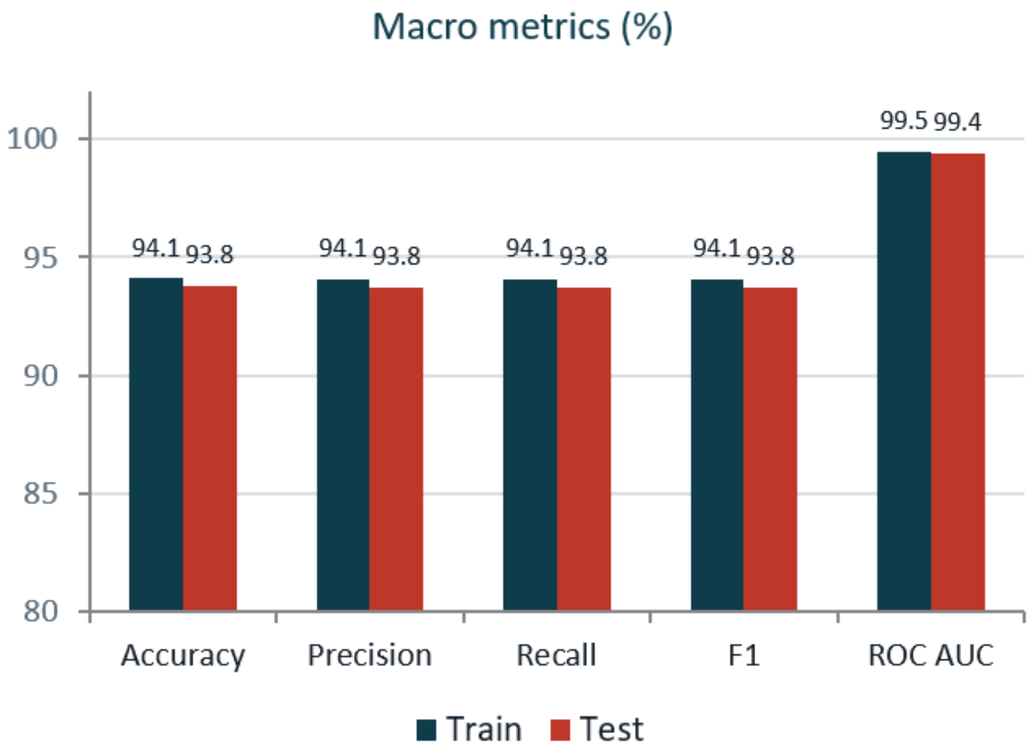
pandas 3.0.6

numpy 2.5.4

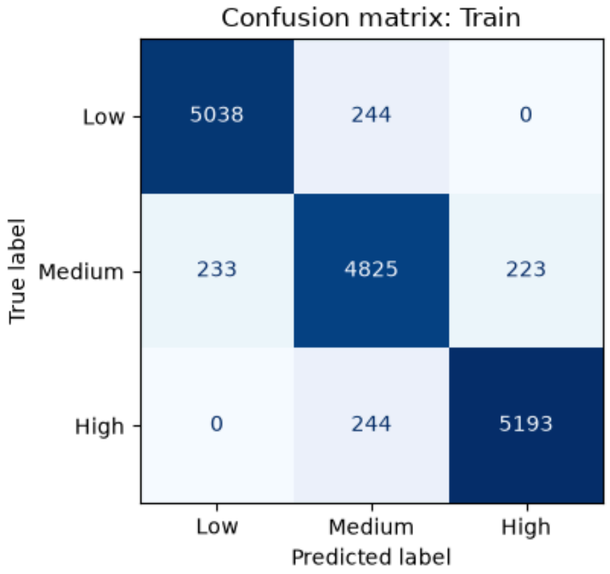
matplotlib 3.11.2

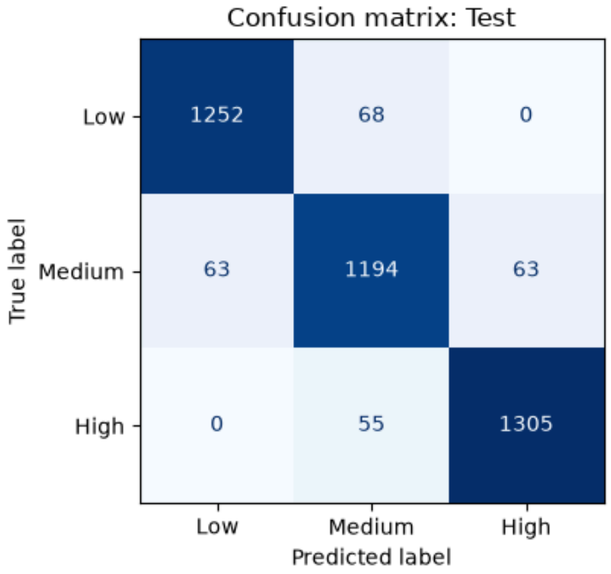
scikit-learn 1.9.1

Results: 93.8% test accuracy, no overfitting



Train n = 16,000, test n = 4,000. Axis starts at 80%.





93.8%

test accuracy

0.994

ROC AUC on the test set

Learnings: what worked and what didn't

WHAT WORKED

- Train 94.1% vs test 93.8%: no overfitting
- Routine tasks, automation and job level drive risk
- Low and High never confused

WHAT DIDN'T

- Medium is hardest to classify(F1=0.906) compared to High & Low (0.957 & 0.950) respectively
- SGD curve plateaus near 92.7% test accuracy, below the final model
- Synthetic data may not match real workplaces

WHO DID WHAT

- Shreyas: GitHub Collaboration, documentation, README
- Hisham: Setup, execution of code, requirements.txt
- Saniya: Slides
- Bhakti: Video
- All Members: Coding and Implementation

AI-use Disclosure

Claude: First draft of README, assisted with code for plotting curves & debugged an error in SGDCclassifier code
No AI was used to generate the dataset. All reported metrics come from running the notebook